

Multi-VLAN Aruba Switch Project (Hands-On)

Documentation by: Rooble Dahir

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Project Overview

This document provides detailed guidelines for configuring DHCP on an Aruba 6300 switch to establish a local network for three clients. The setup includes creating three VLANs, each assigned to a client, with DHCP dynamically distributing IP addresses from a specified range. Additionally, all clients must have internet access and be able to communicate with one another through the switch. Communication between the clients will be secured using SSH with key-based authentication.

Key word definitions for this project

VLAN: Virtual Local Area Network(s) which are used to segment the physical network. We use this to create different networks for each end user (Our PC).

SSH: Secure shell. We use this to access our accounts from one VM to another using our generated key.

IP Routes: Defines a path that network traffic can travel to. We use this to allow our VM's to communicate to each other since we are in three separate VLANs.

DHCP: Dynamic Host Configuration Protocol. This assigns an IP automatically to whomever is at the end. We use this to get an automatic IP that we can use for our VM's.

Subnet Determination for Given Student IDs

In this project, the subnet is defined as **172.16.x.*/26**, where **x** represents the unique ID assigned by random. However, determining the correct subnet involves more than simply replacing **x** with the ID. Instead, we must calculate the appropriate values using the **/26** subnet mask.

Each IP address consists of 32 bits, with **172.16** being fixed and occupying the first **16 bits**. The **/26** subnet mask leaves us with **10 bits** to define our subnet.

To determine the subnet for a given student ID:

1. Convert the **student ID** to a **10-bit binary representation**.
2. Split it into **8-bit + 2-bit** sections.
3. The first **8 bits** form the third octet of the IP address.
4. The remaining **2 bits** contribute to the starting address in the fourth octet.
5. By setting the last **6 bits** to **0**, we determine the first address in the subnet.

Subnet Calculations

1. Client1 (ID 73) PR

- **Binary:** 0001001001
- **Splitting:** 00010010.01??????
- **Converted to Decimal:** 18.64
- **Subnet:** 172.16.18.64/26

2. Client2 (ID 66)

- **Binary:** 0001000010
- **Splitting:** 00010000.10??????
- **Converted to Decimal:** 16.128
- **Subnet:** 172.16.16.128/26

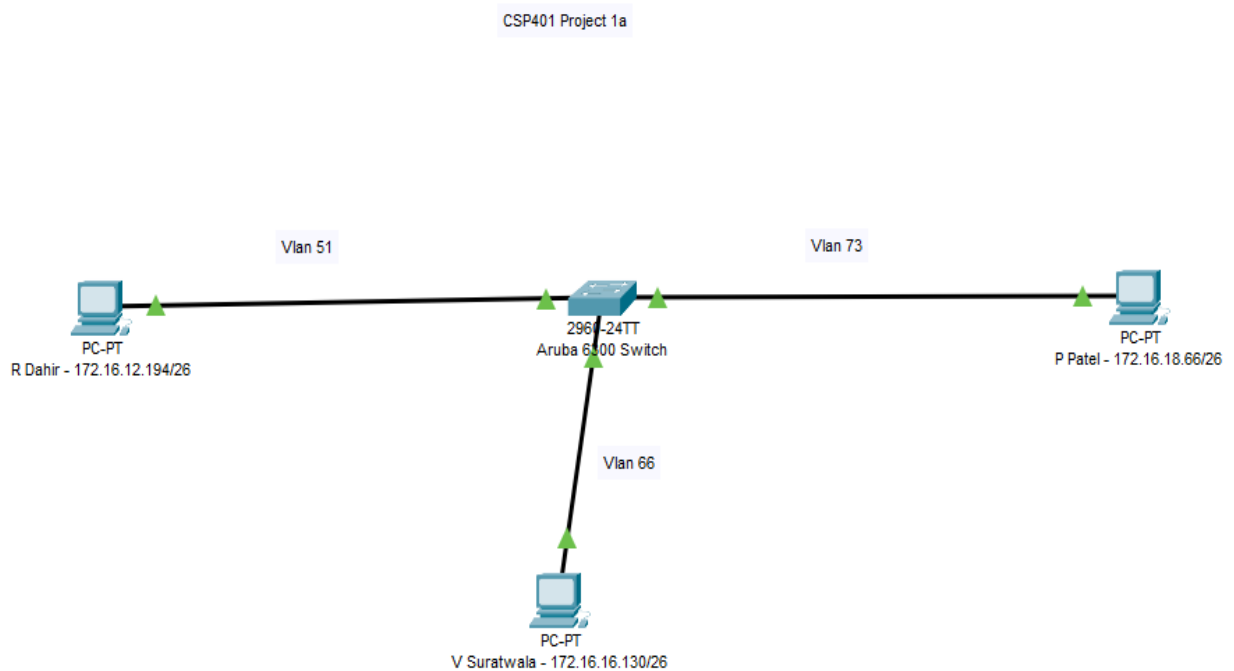
3. Client3 (ID 51)

- **Binary:** 0000110011
- **Splitting:** 00001100.11??????
- **Converted to Decimal:** 12.192
- **Subnet:** 172.16.12.192/26

Thus, the assigned subnets are:

- **Client1:** 172.16.18.64/26
- **Client2:** 172.16.16.128/26
- **Client3:** 172.16.12.192/26

Network Topology



Implementation

Step 1: Setup the physical topology

The first thing done when trying to connect to the switch is making sure everything is plugged in to the correct ports, and that we can connect to the switch. Some things done were:

- Knowing which port was being used for on our computer (G1, G4 etc.). Need to make sure that our port number aligns with the port of the switch. If we didn't have the correct port number, we could not have access to the switch.
- Setting up cables in the back. To make sure that our switch can communicate to our personal computers and VM's, we need to connect the cables into the correct ports.

Step 2: Access To The Switch

The second thing is to make sure we can SSH from the computer to the switch. This can be done by using the command: `ssh <name>@<switch-ip-address>`. After met with the password prompt, no password is needed, and all we need to do is press enter. Now in User Executive Mode.

Step 3: VLAN Configuration

The network topology we were provided with showed that we needed to have three separate VLANs for our three VMs. This is how it was done:

- In configuration mode, using the command “`vlan <number>`” creates a VLAN for you. For us we created 3 different VLANs.
- Since the users within this VLAN are going to be communicating with users outside the VLAN, it needs to have a default gateway.
- Each VLAN was assigned to a specific interface, for example interface 1/1/1 was assigned to one VLAN, interface 1/1/2 was assigned to another VLAN etc.

Step 4: Set-up DHCP

Since static IP addresses are not being used in this project, we need to set up Dynamic Host Configuration Protocol (DHCP). This can allow for automatic IP address assignment instead of manually assigning an IP address. We also need to define the pools of DHCP since when the DHCP server automatically assigns an IP, it picks from a pool of available IP's. To do this:

- Use the command “`dhcp-server vrf default`” to enable DHCP on the switch. It also uses the default VRF instead of a custom one.

- Define the pool. By doing this, you're assigning a range of IP addresses that are available to users of the network.

Step 5: Confirmation On Clients

This step is needed in order to ensure that we are using DHCP, and our VMs can access the internet:

- Each VM should have one adapter set in "Bridged" connecting to the switch. The second adapter should be connected straight to the internet.
- Ensure that the Network adapters are configured correctly. An IP should be obtained automatically, and the `ip a` command should show us the correct IP address.

Step 6: IP Routes And SSH

IP Routes are needed in order to define a path to a specific network. In this case, we need to route traffic to each VM's network in order for them to communicate. This can allow us to SSH.

- Setup IP Routes for each VM, this can be done in the CLI using "`sudo ip route add`" command. There should be two ip route commands for each VM.
- Test ping to ensure that the IP Routes were inserted correctly.

Secure Shell (SSH) is now needed in order to access the other VM's. Users are also needed so that each VM can access a designated account.

- Install OpenSSH server using the command: `sudo apt install openssh-server`
- Each VM should create two users, one for each VM that is going to use SSH. This can be done using the `useradd` command.
- Each VM should also create a keypair on the new user which is essential for connecting to the client VM's, using the command `ssh-keygen`. Then copy the key to

the user with the command: `ssh-copy-id -I [location of public key] [username]@[ip address]`

Step 7: Testing Network Configuration

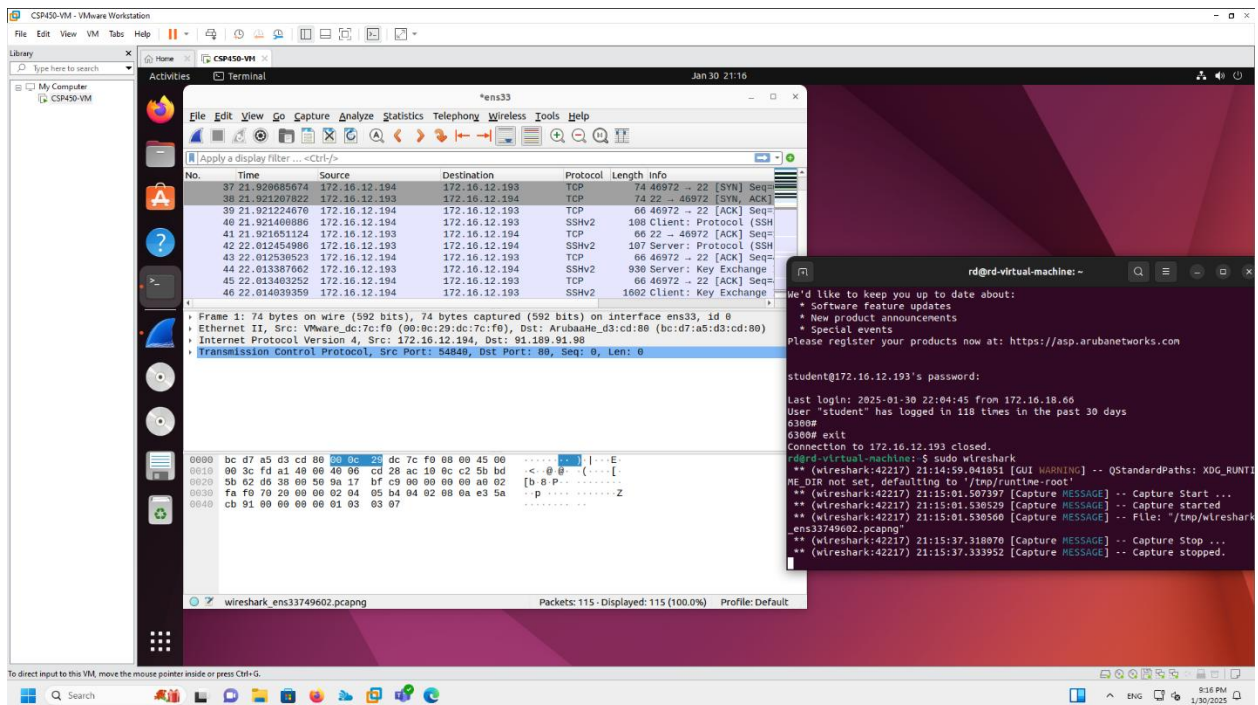
Now that everything is done, we can test all our configuration and ensure that everything works properly.

- Ensure that each VM can be pinged successfully (VM1 to VM2, VM2 to VM3 etc.).
- If you can ping each VM successfully, attempt SSH into each VM. A successful attempt would mean that you can SSH without needing a password.
- Also attempt to SSH into the root account of the user. If you are denied, then that means it was successful.

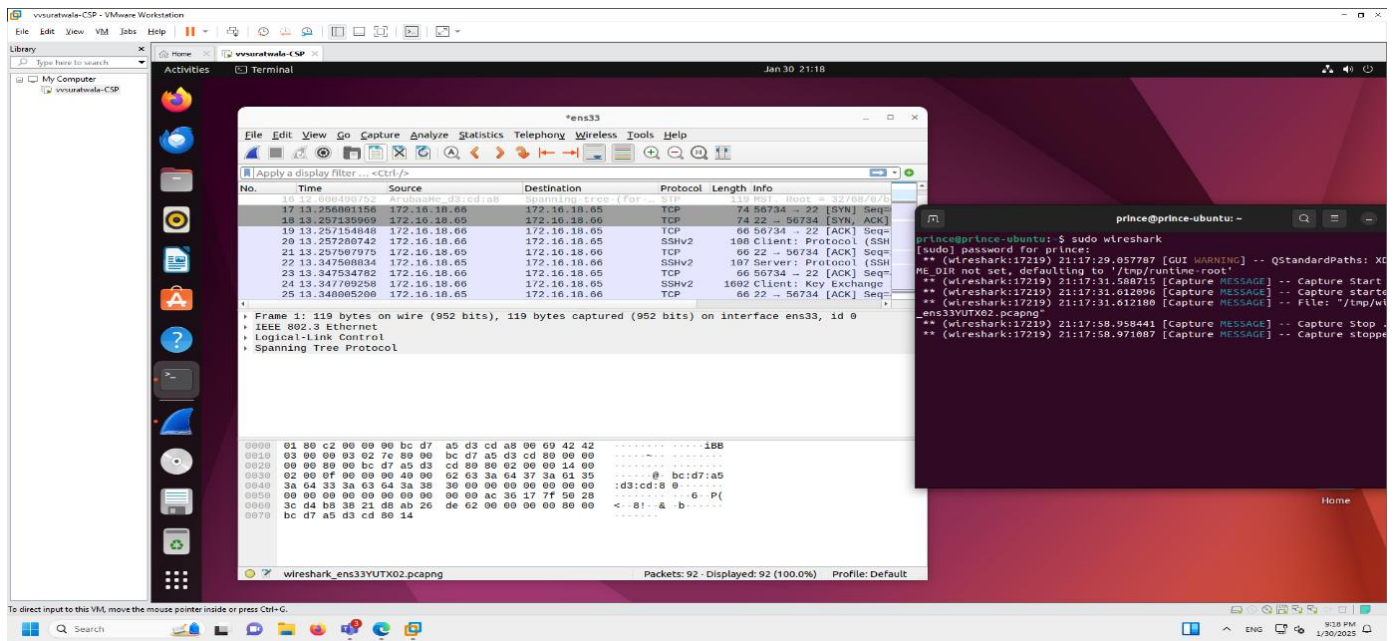
Appendix A: Wireshark

Rooble Dahir: 172.16.12.194 | Prince Patel: 172.16.18.66 | Vrisha Suratwala: 172.16.16.130

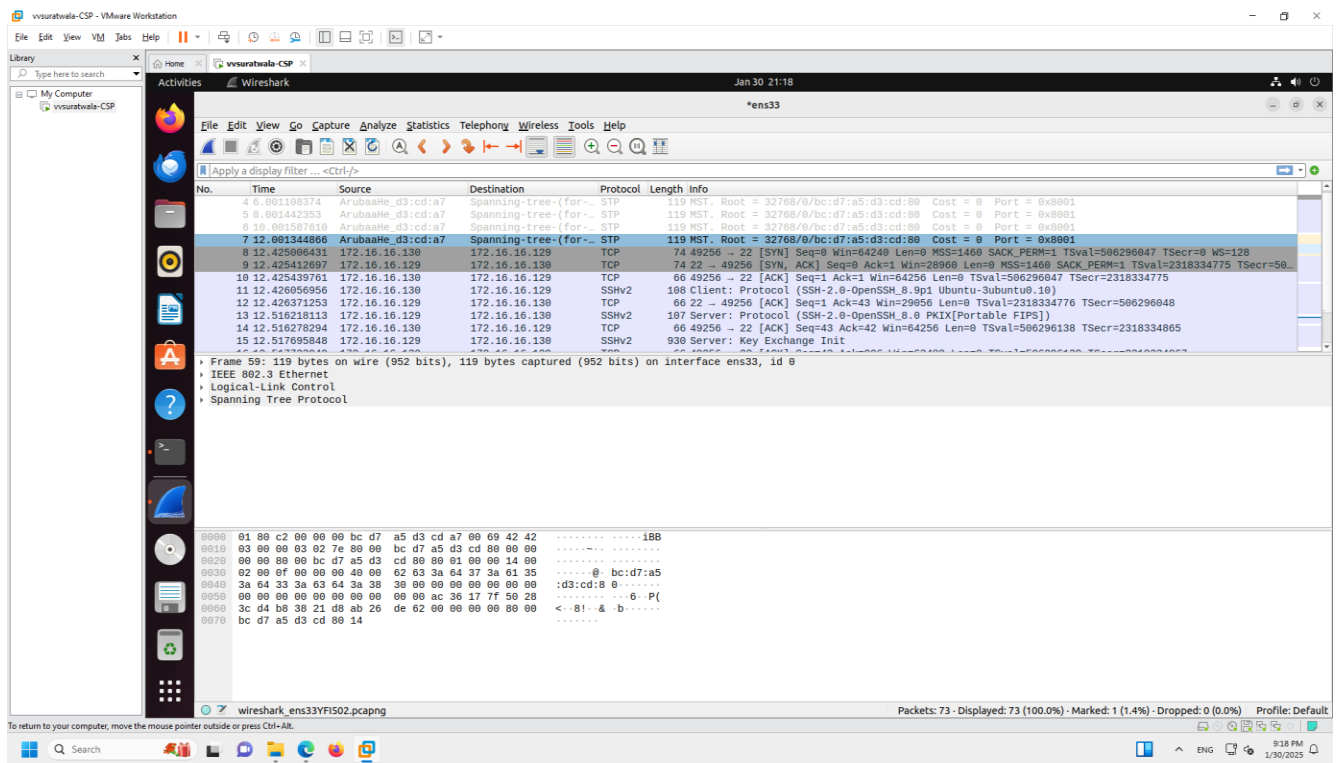
Client3 VM to Switch



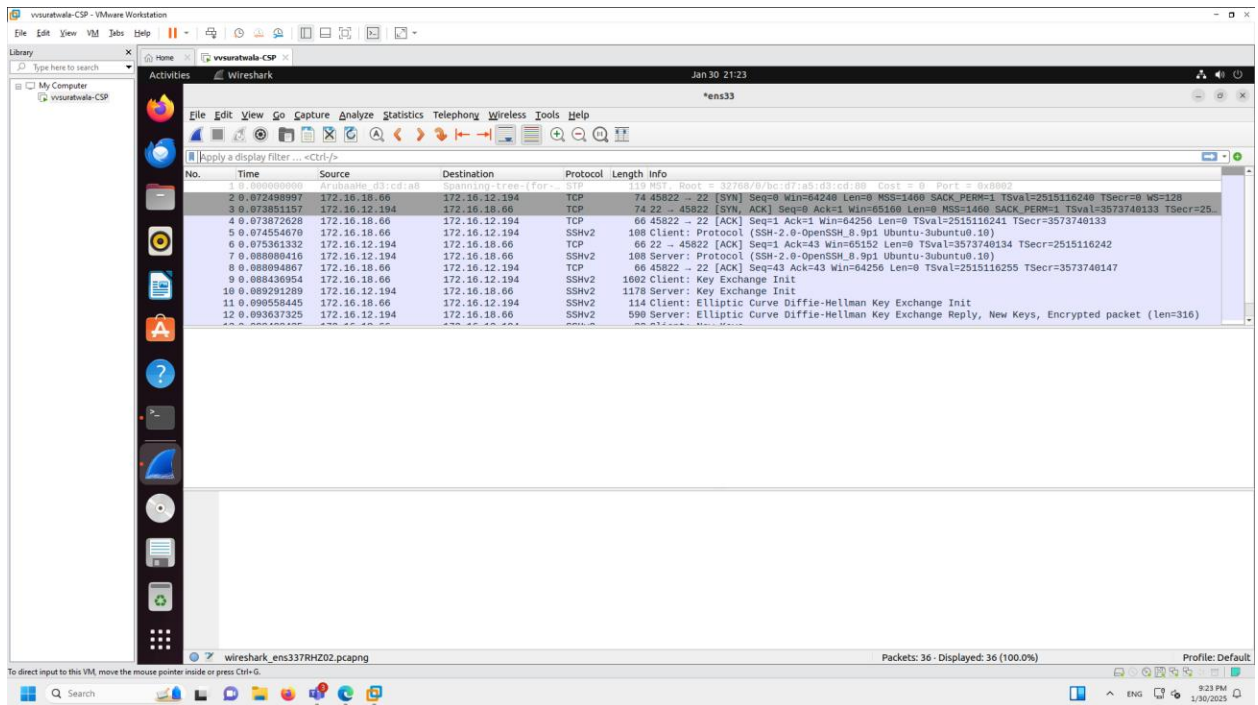
Client1 VM to Switch



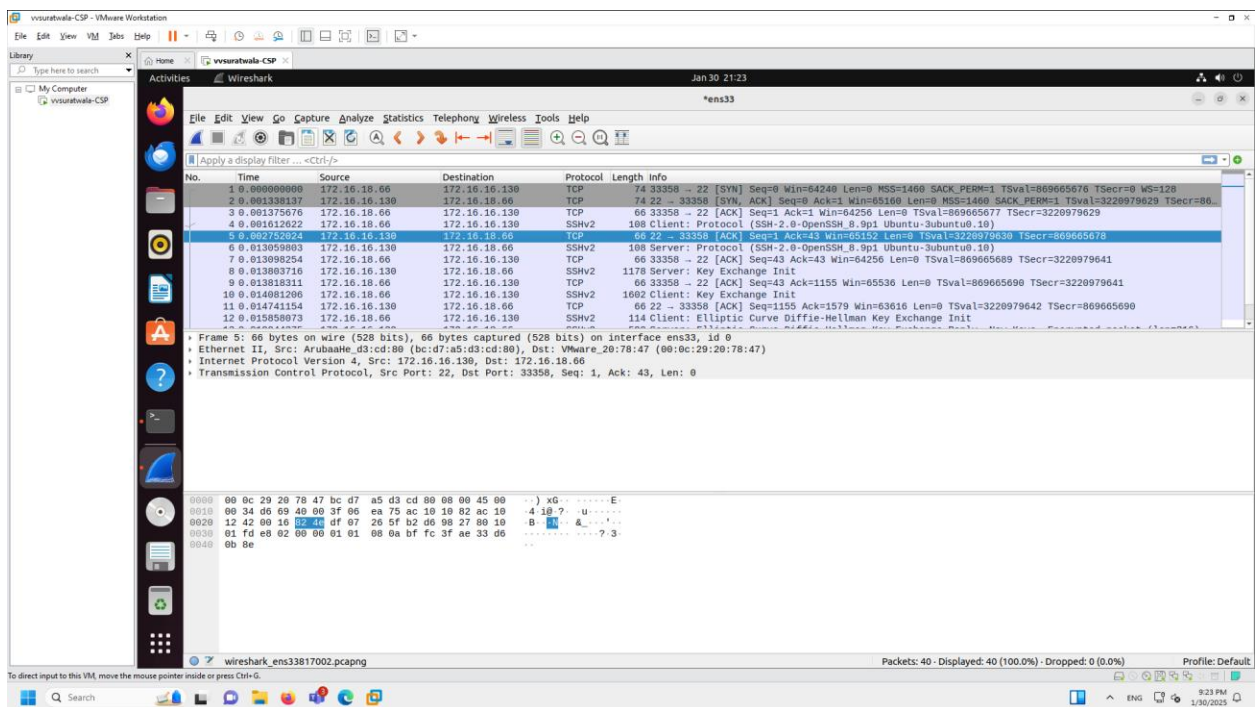
Client2 VM to Switch



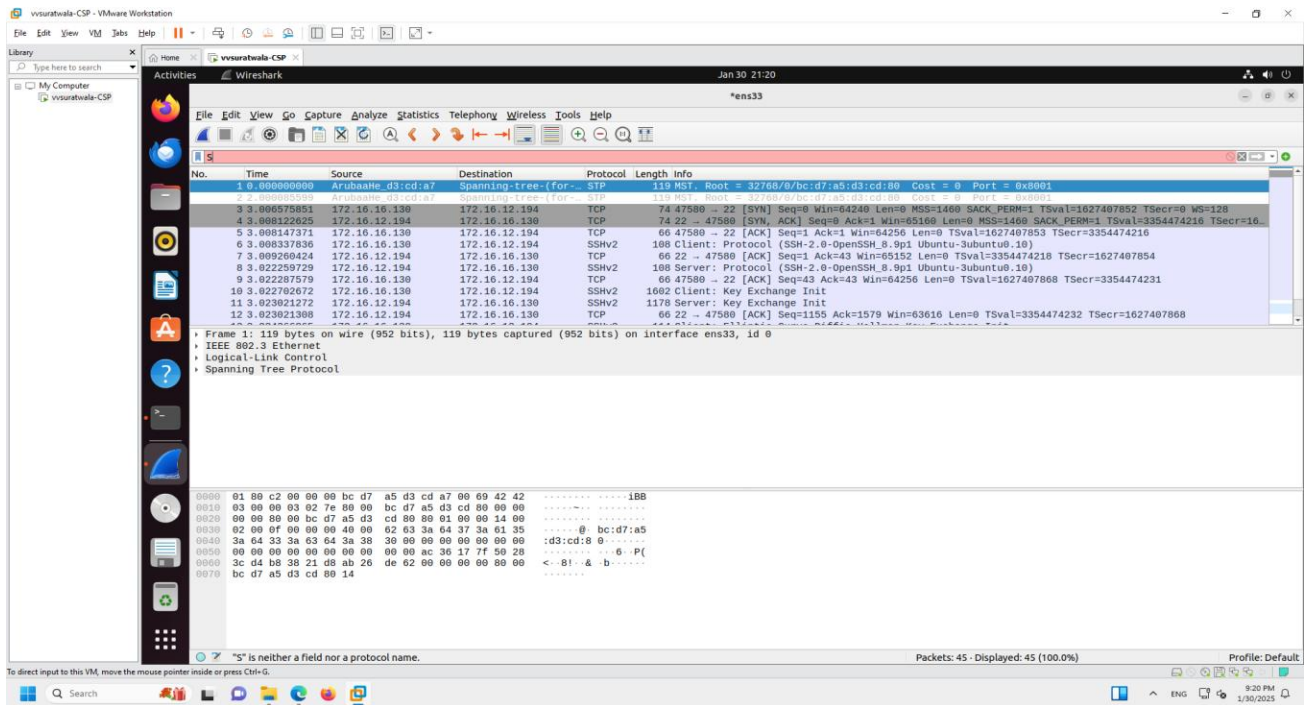
Client3 VM to Client1 VM



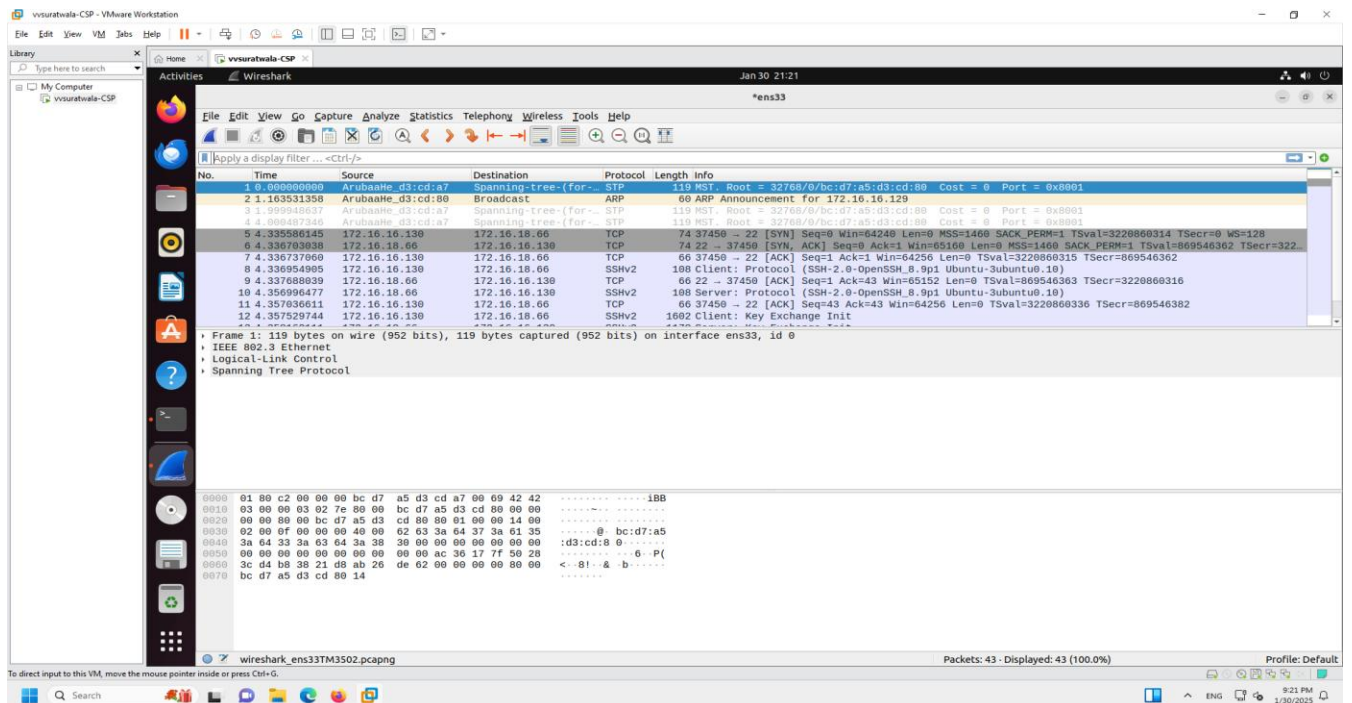
Client1 VM to Client2 VM



Client2 VM to Client1 VM

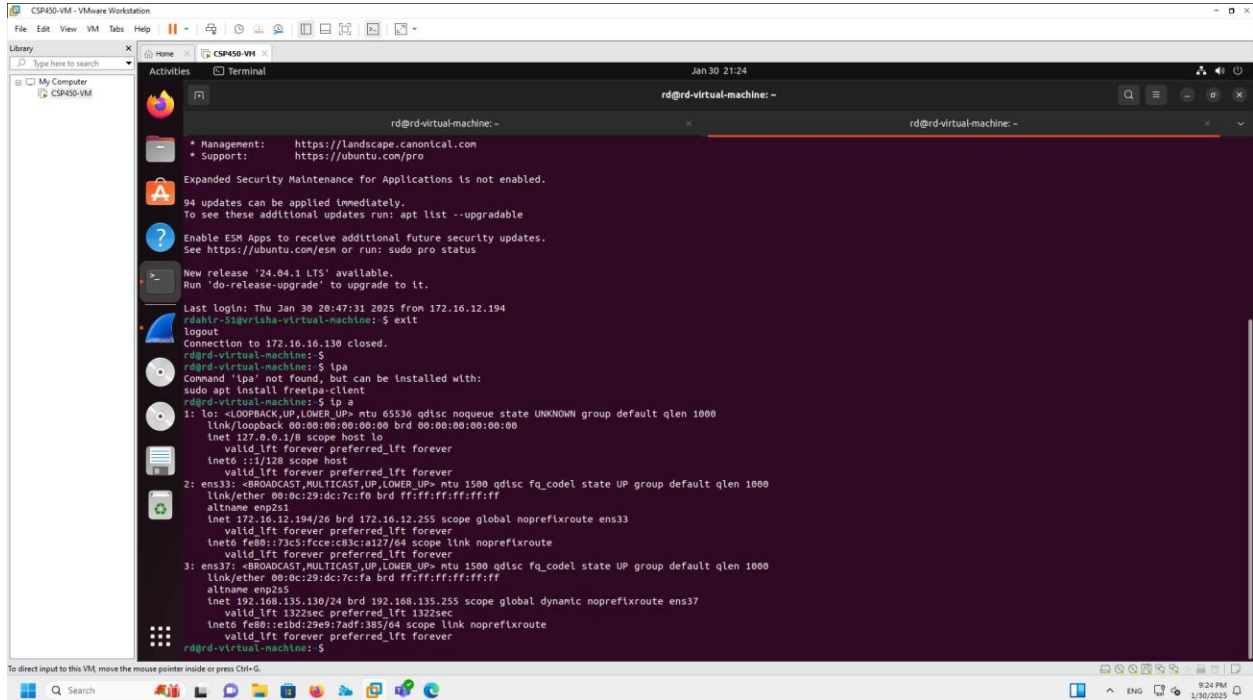


Client2 VM to Client3 VM



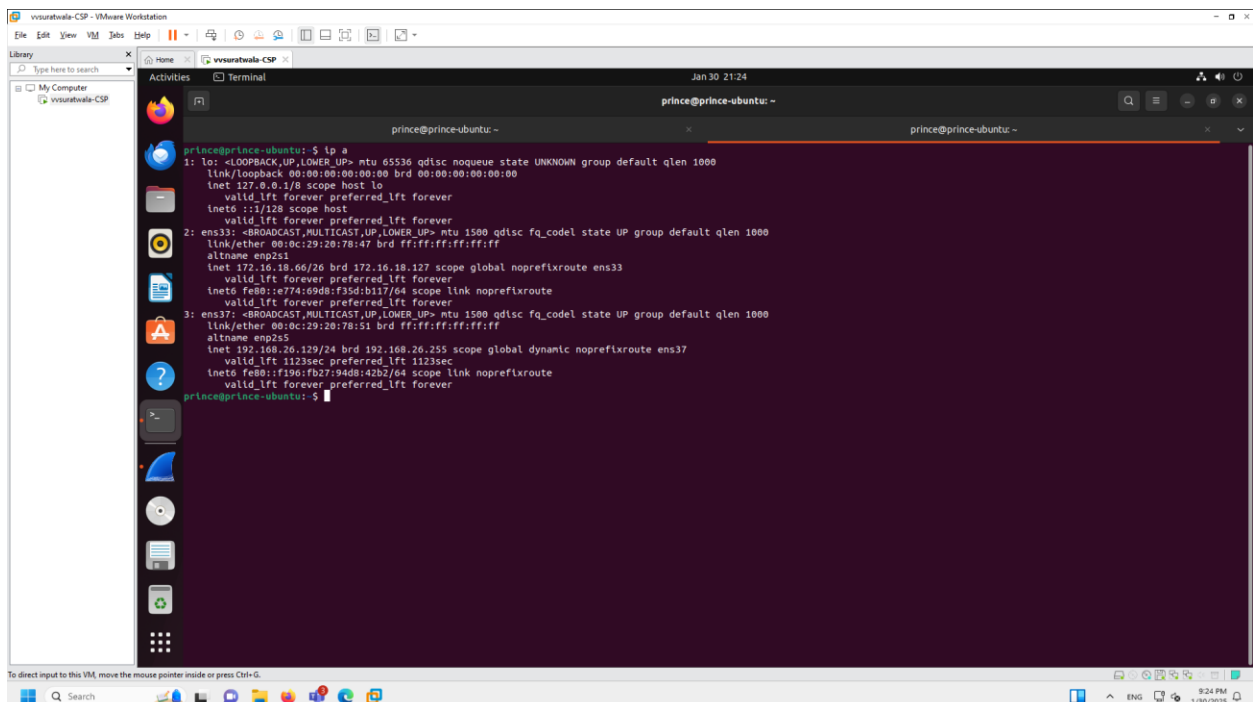
Appendix B: Commands on VM

Terminal Command: ip a Client1 VM



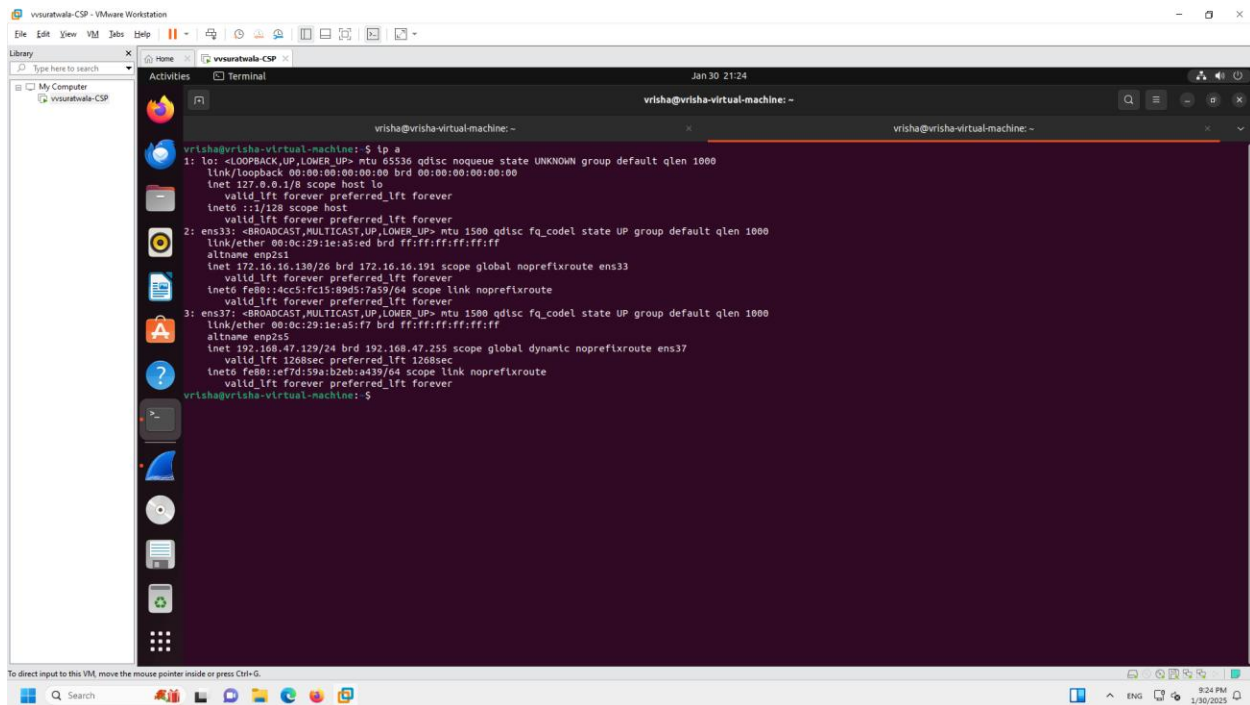
```
rd@rd-virtual-machine: ~  
* Management: https://landscape.canonical.com  
* Support: https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
94 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
New release '24.04.1 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Thu Jan 30 20:47:31 2025 from 172.16.12.194  
rdahlr-51@vrisha-virtual-machine: $ exit  
logout  
Connection to 172.16.130 closed.  
rd@rd-virtual-machine: $  
rd@rd-virtual-machine: $ ip a  
Command 'ip' not found, but can be installed with:  
sudo apt install freempa-client  
rd@rd-virtual-machine: $ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:dc:7c:f0 brd ff:ff:ff:ff:ff:ff  
    altname enp251  
    inet 172.16.12.194/26 brd 172.16.12.255 scope global noprefixroute ens33  
        valid_lft forever preferred_lft forever  
    inet6 fe80::73c5:fcce:c83c:a127/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:dc:7c:fa brd ff:ff:ff:ff:ff:ff  
    altname enp255  
    inet 192.168.135.130/24 brd 192.168.135.255 scope global dynamic noprefixroute ens37  
        valid_lft 1322sec preferred_lft 1322sec  
    inet6 fe80::e1bd:29e9:7adf:385/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
rd@rd-virtual-machine: $
```

Client3 VM



```
prince@prince-ubuntu: ~  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:20:78:47 brd ff:ff:ff:ff:ff:ff  
    altname enp251  
    inet 172.16.18.66/26 brd 172.16.18.127 scope global noprefixroute ens33  
        valid_lft forever preferred_lft forever  
    inet6 fe80::e774:6908:f3d6:b17/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:20:78:51 brd ff:ff:ff:ff:ff:ff  
    altname enp255  
    inet 192.168.26.129/24 brd 192.168.26.255 scope global dynamic noprefixroute ens37  
        valid_lft 1123sec preferred_lft 1123sec  
    inet6 fe80::f196:fd27:94d8:42b2/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
prince@prince-ubuntu: $
```

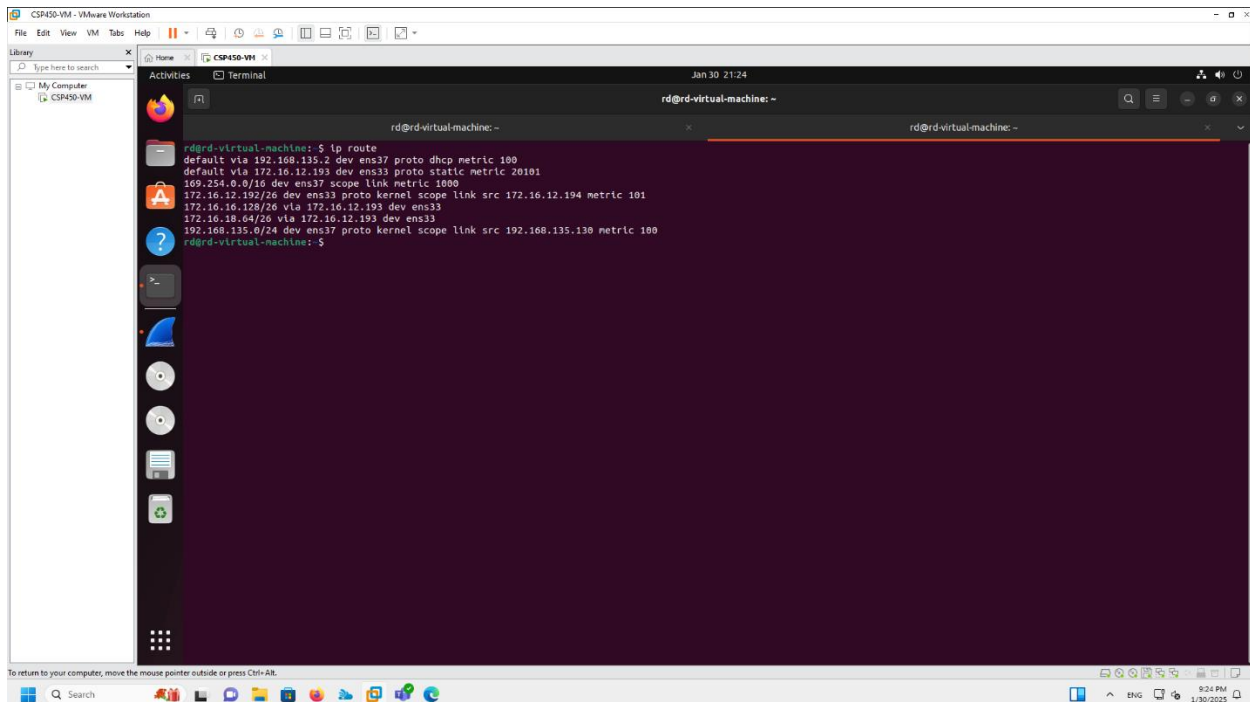
Client2 VM



```
vrisha@vrisha-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1e:a5:ed brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.16.16.138/26 brd 172.16.16.191 scope global noprefixroute ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::4cc5:fc15:89d5:7a59/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1e:a5:f7 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.47.129/24 brd 192.168.47.255 scope global dynamic noprefixroute ens37
        valid_lft 1268sec preferred_lft 1268sec
    inet6 fe80::e7d:59a:b2eb:e439/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
vrisha@vrisha-virtual-machine:~$
```

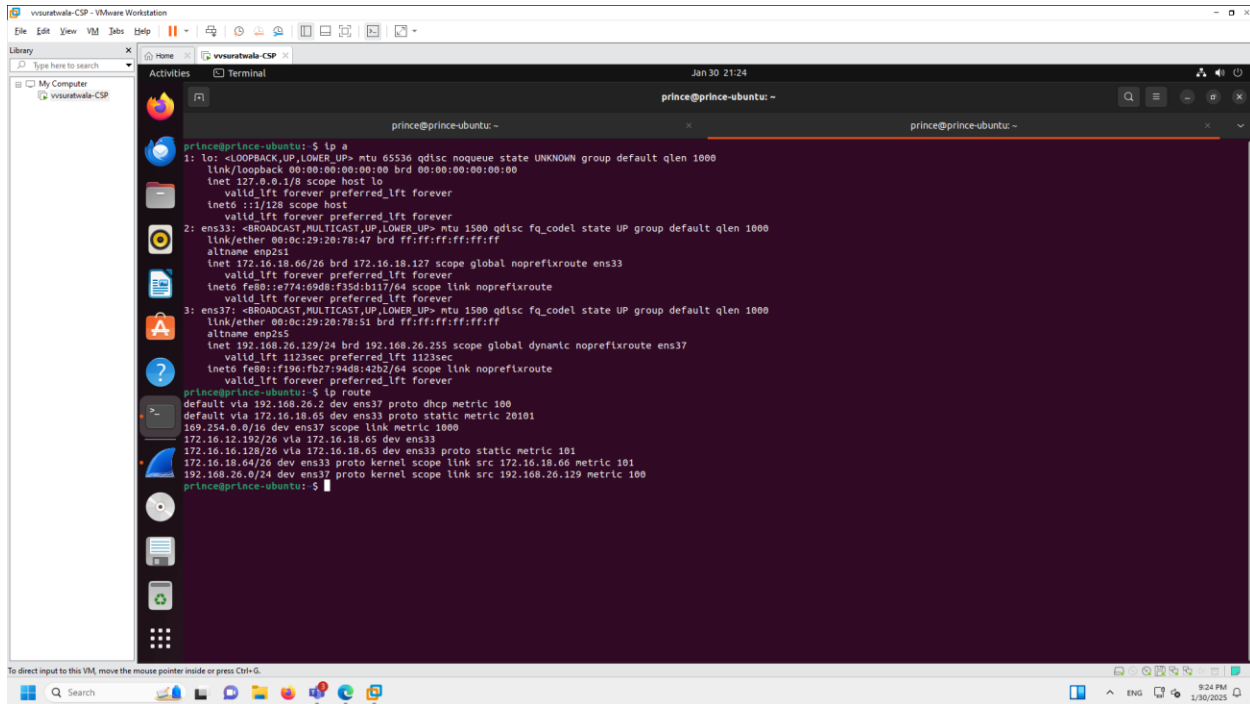
Terminal Command: IP Route

Client1 VM



```
rd@rd-virtual-machine:~$ ip route
default via 192.168.135.2 dev ens37 proto dhcp metric 100
default via 172.16.12.193 dev ens33 proto static metric 20101
169.254.0.0/16 dev ens37 scope link metric 1000
172.16.12.192/26 dev ens33 proto kernel scope link src 172.16.12.194 metric 101
172.16.16.128/26 via 172.16.12.193 dev ens33
172.16.18.0/24 via 172.16.12.193 dev ens33
192.168.135.0/24 dev ens37 proto kernel scope link src 192.168.135.130 metric 100
rd@rd-virtual-machine:~$
```

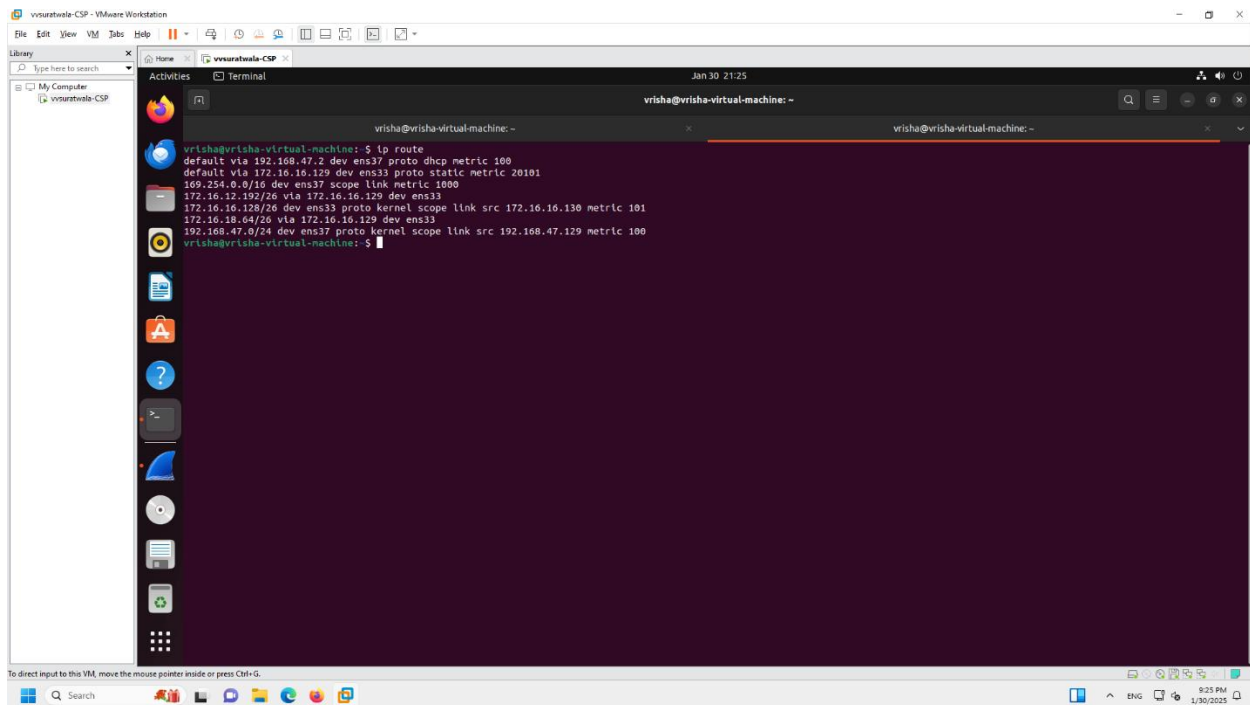
Client3 VM



prince@prince-ubuntu: ~
prince@prince-ubuntu: ~
prince@prince-ubuntu: ~

```
prince@prince-ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:20:78:51 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.16.18.66/26 brd 172.16.18.127 scope global noprefixroute ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::e774:69d8:f35d:b117/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:20:78:51 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.26.129/24 brd 192.168.26.255 scope global dynamic noprefixroute ens37
        valid_lft 1123sec preferred_lft 1123sec
    inet6 fe80::f360:f027:94d8:42b7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
prince@prince-ubuntu:~$ ip route
default via 192.168.26.2 dev ens37 proto dhcp metric 100
default via 172.16.18.65 dev ens33 proto static metric 20101
169.254.0.0/16 dev ens37 scope link metric 1000
172.16.12.192/26 via 172.16.18.65 dev ens33
172.16.16.128/26 via 172.16.18.65 dev ens33 proto static metric 101
172.16.18.64/26 dev ens33 proto kernel scope link src 172.16.18.66 metric 101
192.168.26.0/24 dev ens37 proto kernel scope link src 192.168.26.129 metric 100
prince@prince-ubuntu:~$
```

Client2 VM

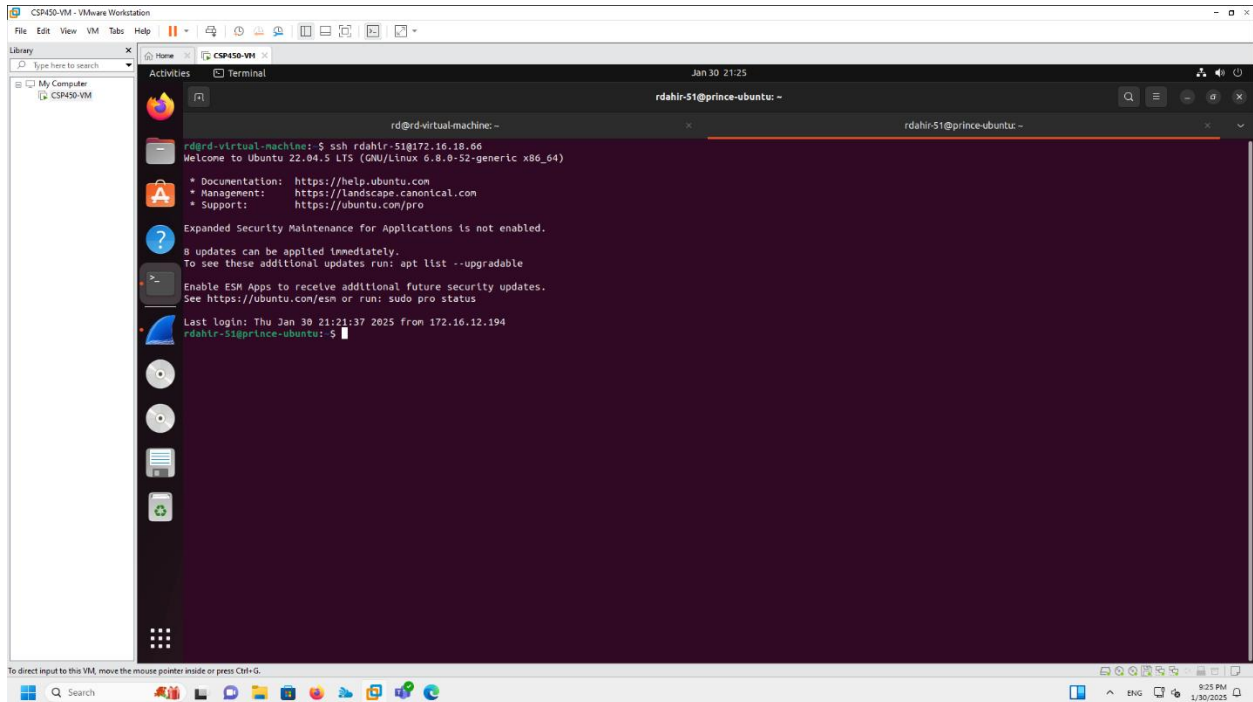


vrisha@vrisha-virtual-machine: ~
vrisha@vrisha-virtual-machine: ~
vrisha@vrisha-virtual-machine: ~

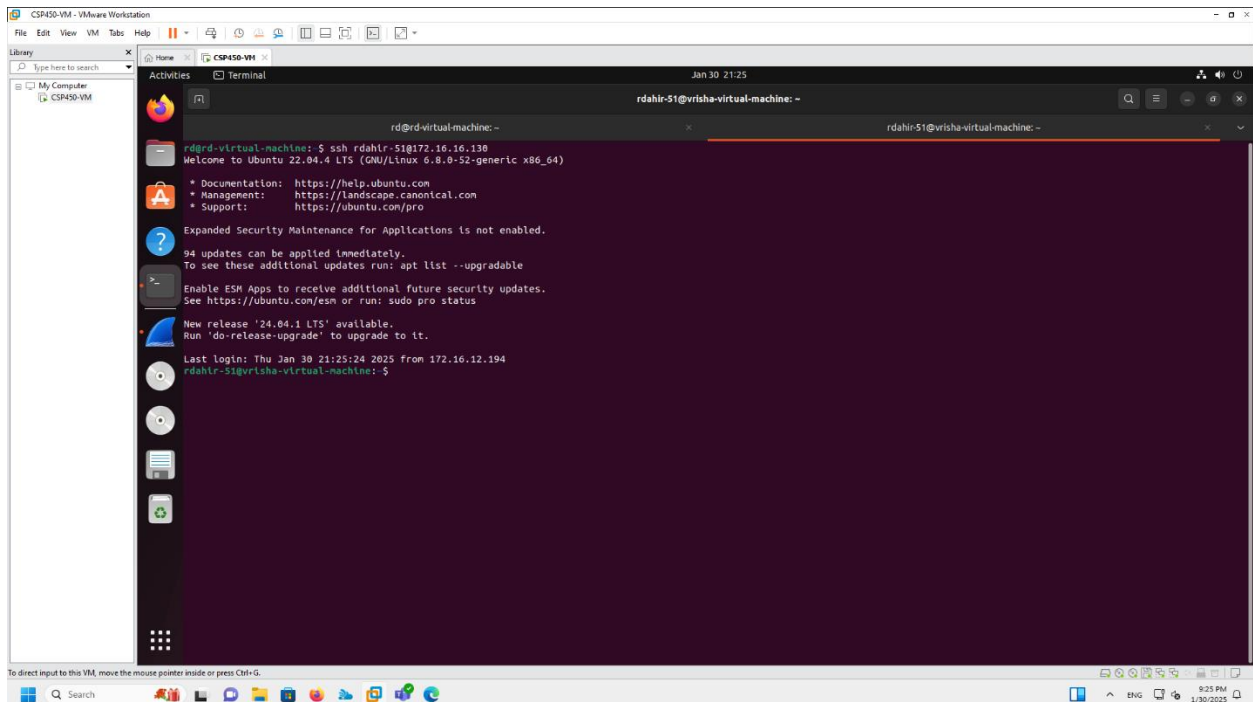
```
vrisha@vrisha-virtual-machine:~$ ip route
default via 192.168.47.2 dev ens37 proto dhcp metric 100
default via 172.16.18.129 dev ens33 proto static metric 20101
169.254.0.0/16 dev ens37 scope link metric 1000
172.16.12.192/26 via 172.16.16.129 dev ens33
172.16.16.128/26 dev ens33 proto kernel scope link src 172.16.16.130 metric 101
172.16.18.64/26 via 172.16.16.129 dev ens33
192.168.47.0/24 dev ens37 proto kernel scope link src 192.168.47.129 metric 100
vrisha@vrisha-virtual-machine:~$
```


SSH To Partner's

Client1 VM to Client3 VM



Client1 VM to Client2 VM



[illegible]

The screenshot displays the VMware Workstation interface with two virtual machines (VMs) running Ubuntu. The left VM, named 'vvsuratale-CSP', is in the 'Activities' view, showing a terminal window with the following output:

```
prince@prince-ubuntu: ~
172.16.18.64/26 dev ens33 proto kernel scope link src 172.16.18.66 metric 101
192.168.26.0/24 dev ens37 proto kernel scope link src 192.168.26.129 metric 100
prince@prince-ubuntu: ~$ ssh ppatel-73@172.16.12.194
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-52-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan 30 21:23:44 2025 from 172.16.18.66
ppatel-73@rd-virtual-machine: ~$
ppatel-73@rd-virtual-machine: ~$ exit
logout
Connection to 172.16.12.194 closed.
prince@prince-ubuntu: ~$ ssh ppatel-73@172.16.12.194
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.8.0-52-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

94 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

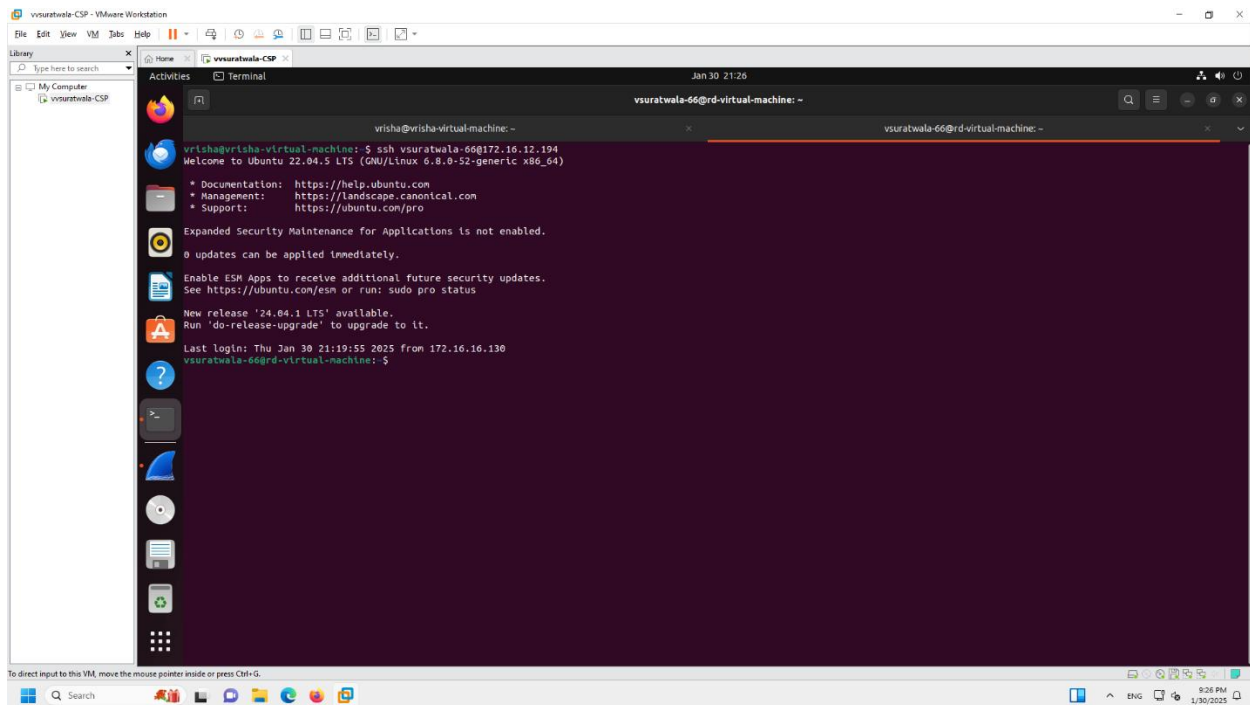
Last login: Thu Jan 30 21:22:58 2025 from 172.16.18.66
ppatel-73@rd-virtual-machine: ~$
```

The right VM, also named 'vvsuratale-CSP', is in the 'Terminal' view, showing a terminal window with the following output:

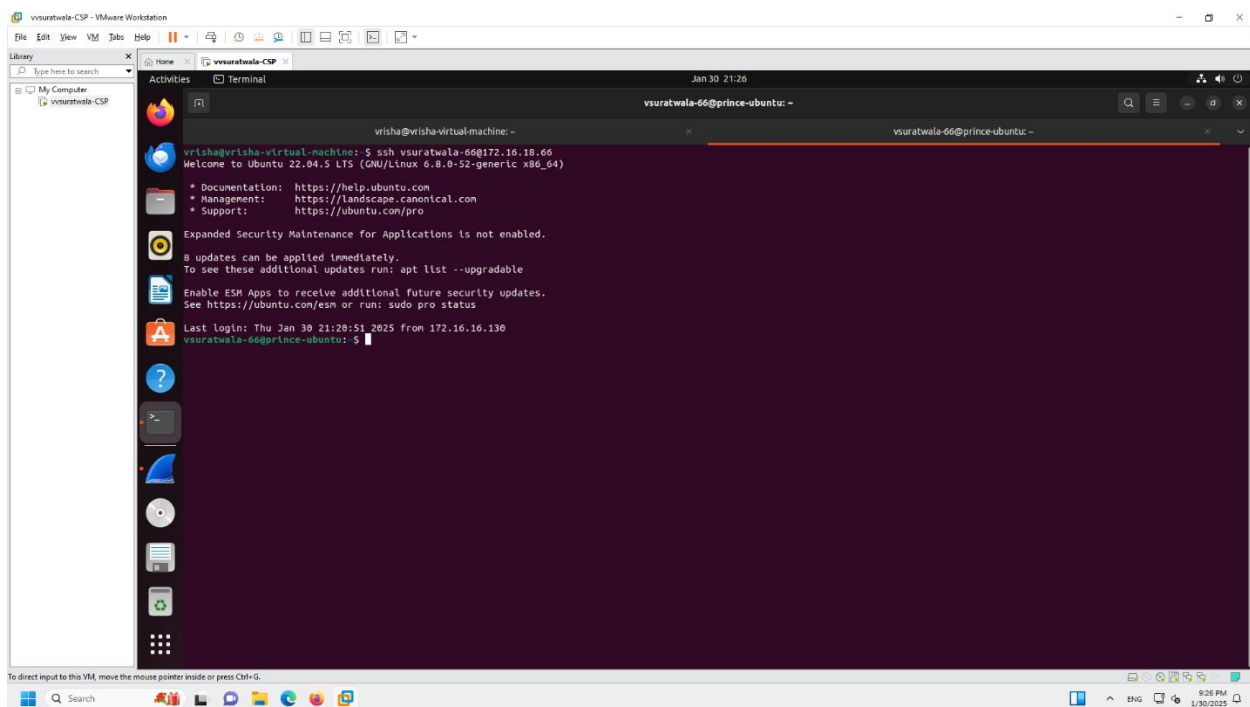
```
ppatel-73@vrisa-virtual-machine: ~
ppatel-73@vrisa-virtual-machine: ~$
```

The interface includes a top menu bar with options like File, Edit, View, VM, Tools, Help. A left sidebar shows a library with a search bar and a list of VMs. The bottom status bar indicates the current input device and provides instructions for direct input to the VM.

Client2 VM to Client1 VM

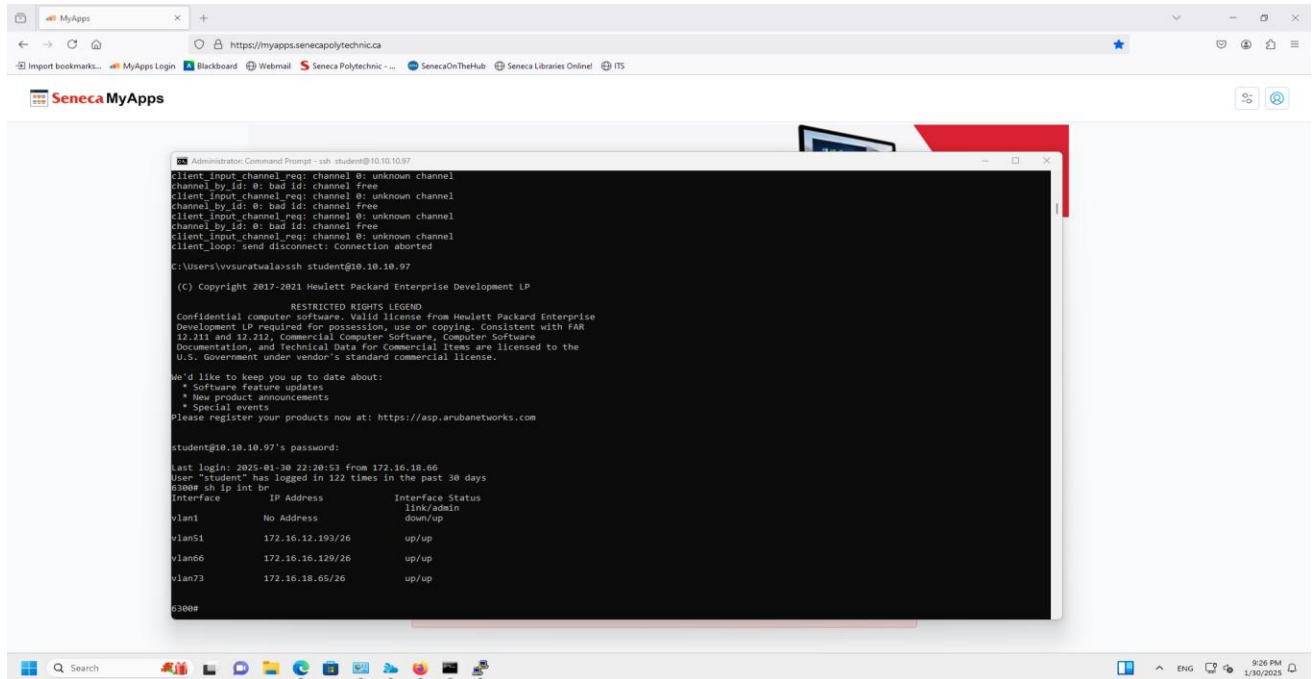


Client2 VM to Client3 VM

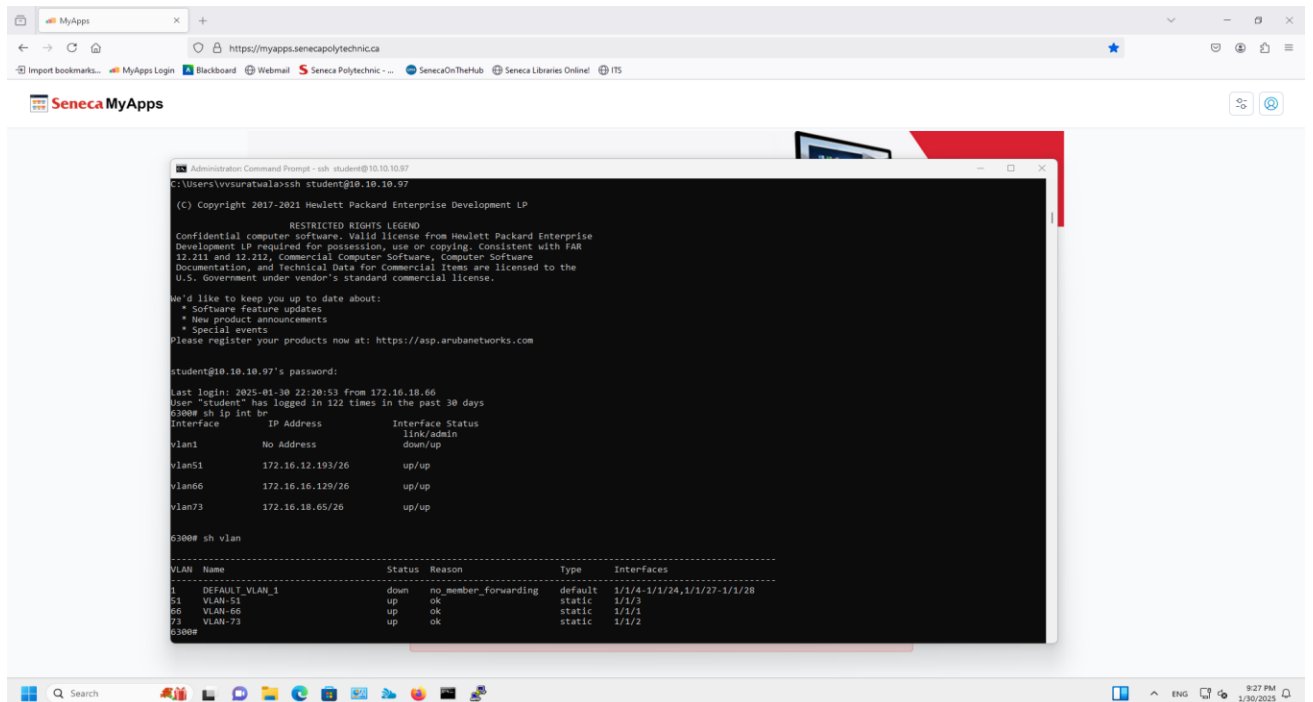


Appendix C: Commands on Switch

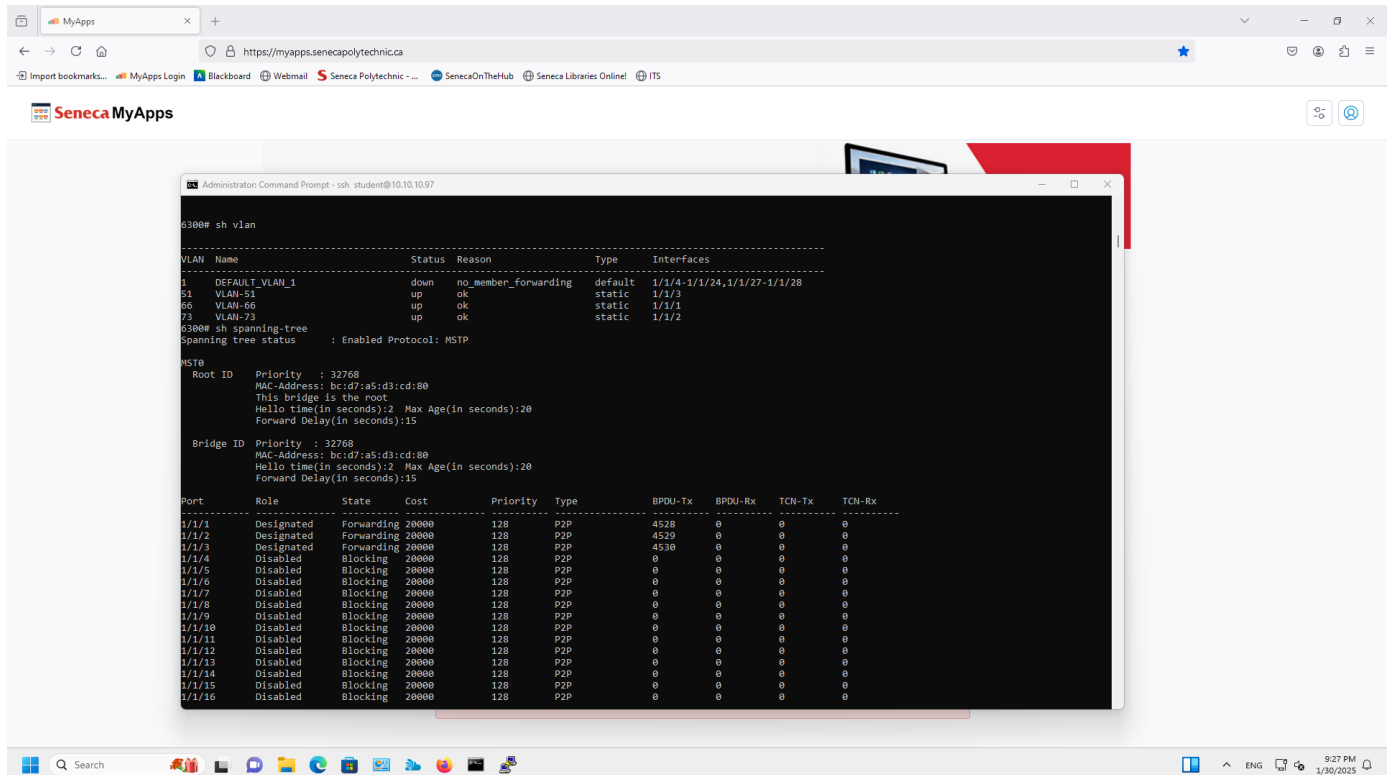
Sh ip int br



Sh vlan



Sh spanning-tree



```
Administrator: Command Prompt - ssh student@10.10.10.97

6300# sh vlan

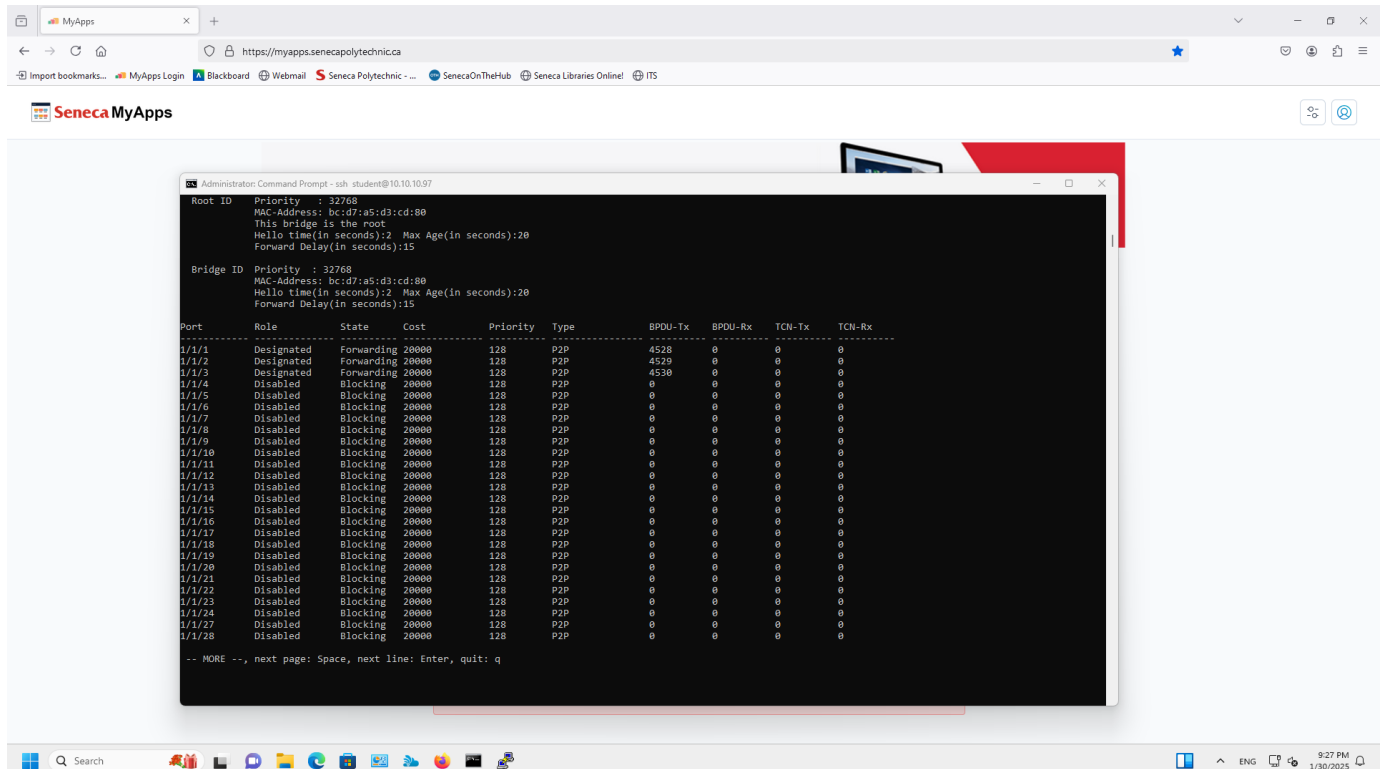
VLAN Name                Status Reason                Type    Interfaces
-----
1    DEFAULT_VLAN_1         down  no_member_forwarding      default 1/1/4-1/1/24,1/1/27-1/1/28
51   VLAN-51                up    ok                        static  1/1/3
66   VLAN-66                up    ok                        static  1/1/1
73   VLAN-73                up    ok                        static  1/1/2

6300# sh spanning-tree
spanning tree status      : Enabled Protocol: MSTP

MST0
  Root ID    Priority : 32768
            MAC-Address: bc:d7:a5:d3:cd:80
            This bridge is the root
            Hello time(in seconds):2 Max Age(in seconds):20
            Forward Delay(in seconds):15

  Bridge ID  Priority : 32768
            MAC-Address: bc:d7:a5:d3:cd:80
            Hello time(in seconds):2 Max Age(in seconds):20
            Forward Delay(in seconds):15

Port    Role    State    Cost    Priority Type    BPDU-Tx    BPDU-Rx    TCN-Tx    TCN-Rx
-----
1/1/1   Designated Forwarding 20000    128     P2P     4528       0          0         0
1/1/2   Designated Forwarding 20000    128     P2P     4529       0          0         0
1/1/3   Designated Forwarding 20000    128     P2P     4530       0          0         0
1/1/4   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/5   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/6   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/7   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/8   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/9   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/10  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/11  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/12  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/13  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/14  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/15  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/16  Disabled Blocking 20000    128     P2P     0          0          0         0
```



```
Administrator: Command Prompt - ssh student@10.10.10.97

Root ID    Priority : 32768
MAC-Address: bc:d7:a5:d3:cd:80
This bridge is the root
Hello time(in seconds):2 Max Age(in seconds):20
Forward Delay(in seconds):15

Bridge ID  Priority : 32768
MAC-Address: bc:d7:a5:d3:cd:80
Hello time(in seconds):2 Max Age(in seconds):20
Forward Delay(in seconds):15

Port    Role    State    Cost    Priority Type    BPDU-Tx    BPDU-Rx    TCN-Tx    TCN-Rx
-----
1/1/1   Designated Forwarding 20000    128     P2P     4528       0          0         0
1/1/2   Designated Forwarding 20000    128     P2P     4529       0          0         0
1/1/3   Designated Forwarding 20000    128     P2P     4530       0          0         0
1/1/4   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/5   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/6   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/7   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/8   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/9   Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/10  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/11  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/12  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/13  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/14  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/15  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/16  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/17  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/18  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/19  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/20  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/21  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/22  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/23  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/24  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/27  Disabled Blocking 20000    128     P2P     0          0          0         0
1/1/28  Disabled Blocking 20000    128     P2P     0          0          0         0

-- MORE --, next page: Space, next line: Enter, quit: q
```

Sh dhcp-server leases

```
10.10.10.97 - PuTTY
6300# show dhcp-server leases
IP-Address      Client-Id      Expiry-Time    Client-Hostname VRF-Name      Link-Address
-----
172.16.12.223    01:00:0c:29:fa:34:f3  19:13:20 06/02/2025  rd-virtual-machine  default        00:0c:29:fa:34:f3
172.16.16.175    01:00:0c:29:5d:2d:71  19:16:02 06/02/2025  vrishab-virtual-machine  default        00:0c:29:5d:2d:71
172.16.19.73     01:00:0c:29:a0:9a:12  19:57:59 06/02/2025  prince-ubuntu        default        00:0c:29:a0:9a:12
6300#
```

Appendix D: Switch Script Commands

```
conf ter
```

```
vlan 66,73,51  
spanning-tree
```

```
int 1/1/1  
no shutdown  
no routing  
vlan access 51  
exit
```

```
int 1/1/2  
no shutdown  
no routing  
vlan access 73  
exit
```

```
int 1/1/3  
no shutdown  
no routing  
vlan access 66  
exit
```

```
vlan 66  
name VLAN-66  
exit
```

```
vlan 73  
name VLAN-73  
exit
```

```
vlan 51  
name VLAN-51  
exit
```

```
interface vlan 1  
ip dhcp
```

```
interface vlan 51  
ip address 172.16.12.193/26  
no shutdown
```

exit

interface vlan 73
ip address 172.16.18.66/26
no shutdown
exit

interface vlan 66
ip address 172.16.16.130/26
no shutdown
exit

dhcp-server vrf default
pool VLAN-66
range 172.16.16.130 172.16.16.190 prefix-len 26
default-router 172.16.16.129
exit
pool VLAN-51
range 172.16.12.194 17.16.12.254 prefix-len 26
default-router 172.16.12.193
exit
pool VLAN-73
range 172.16.18.66 172.16.18.126
default-router 172.16.18.65
exit
enable